

Data Management Plan

SE498 Team 3 “Audio Rate” · what is collected, where it lives, who sees it, what leaves · state as of 2026-09-16

For future Claude

The operational half of Audio Data Ethics and Handling: session-bundle layout, naming, storage, access, the speech-review step, what the public repo ships, and the retention and deletion promises made in Consent Form - Audio Rate. Everything here follows the hard rules in `CLAUDE.md` (raw audio never in git or the vault; participant identities never in notes). [TBD] = team decision still open. Companion: Lab Session Runbook (produces the bundle), Pipeline Input to Verdict (consumes it).

1. What a session produces

One **session bundle** per participant, one folder, named by code and date: `P07_2026-10-02/`

File	Content	Source
<code>audio_A.wav</code> , <code>audio_B.wav</code> , (<code>audio_C.wav</code>)	16 kHz mono WAV per phone, whole session	capture app
<code>imu_A.csv</code> , ...	accelerometer + gyroscope, audio-clock timestamps	capture app (Decision - IMU logged by the capture app)
<code>h10_rr.csv</code>	RR intervals with BLE arrival and strap timestamps	app, Heart Rate Service
<code>h10_ecg.csv</code> , <code>h10_acc.csv</code>	raw ECG (~130 Hz) and chest accelerometer	app, Polar SDK (Decision - Breathing-rate reference H10 first belt conditional)
<code>events.csv</code>	every event button press with clock time	app
<code>ratings.csv</code>	Borg overall / legs / breathing per stage and per 5 min of the bout	typed from the sheet
<code>metadata.json</code>	session-sheet fields (Lab Session Runbook §6): phones, gain, distances, wattages, room temperature, water, caffeine, sleep, Tanaka HRmax, resting HR, readbacks	typed + app export
<code>qa.json</code>	QA outcomes: cal-tone level, floor, clap offsets, RR rejection rate, lag, breathing-rate check, speech strip list	QA script + reviewer
<code>derived/</code>	log-mel features, labels, windows (regenerable)	pipeline

Nothing in the bundle carries a name. The bundle is **the** unit of QA and of deletion.

2. Where it lives

Copy	Location	Who	Encryption
Capture	the phones, until copied the same day, then deleted from the phones	experimenter	device encryption
Master	encrypted external drive [TBD] (or university-approved encrypted storage [TBD]) held by Josh	Josh + [TBD]	required
Working	the modelling laptop, <code>derived/</code> and bundles for the current experiment only	Josh	full-disk encryption
Never	the vault, GitHub, chat uploads, cloud drives without approval, email	—	—

The **code-to-name key** and the signed consent forms live in a separate physical folder [TBD], never on the same drive as the bundles.

3. Speech review (the consent promise)

Breath audio near the mouth records anything said. Per session, after QA: run voice-activity detection on every audio file → a human listens to each flagged segment → segments with speech are listed in `qa.json` with start/end. Those segments are **excluded from every feature table and released artefact**; the raw WAV is not edited (it stays encrypted). The rating exchanges are speech too — they are stripped the same way, which is why ratings are typed from the sheet, not transcribed.

4. What leaves the master drive

Artefact	Leaves to	Rule
Per-window log-mel features + labels	the repo, as the reproducibility dataset	only after the re-identification audit (Risk 14: Paper - Lu 2020 BreathID breath biometrics from speech, Paper - Tran 2023 Breath sound person identification noisy) — if released features identify participants above chance, release coarser features or none
Model weights, evaluation scripts, session <code>metadata.json</code> (no names), <code>qa.json</code>	the repo	yes
Aggregate plots, ledger rows	this vault, the report	yes
Raw audio, ECG, IMU	nowhere	never; on request to Prof. Dong in person, de-identified, if she asks
Anything	outside the class	only with Prof. Dong's written approval

5. Retention and deletion (as consented)

- Deletion on request until **2026-11-20** (the feature-freeze date; change here and on the form together). A request deletes the bundle, its `derived/`, and its rows from any feature table not yet frozen; the code stays in the key sheet marked *withdrawn*.
- After the freeze, de-identified features may already be in the analysis and cannot be traced back — the form says so.
- Raw audio, ECG and IMU: deleted, or kept only in encrypted archive, within **twelve months of the end of the course** (by 2027-12-31). Owner [TBD].
- Consent forms and the key: kept by [TBD] for the same period, then shredded.

6. Repo layout (reproducibility is graded)

`github.com/cleanpxe/audio-rate` — `capture/` (iOS app) · `features/` (frozen front-end, shared with the app) · `labels/` (RR → %HRR, lag fit) · `model/` · `eval/` (LOPO, LODO, per-phase, zones, baselines) · `data/` (released features + data card only) · `docs/` (this plan, the runbook, the pipeline note as exported PDFs). A **data card** states n, sessions, devices, sample rate, exclusion rules, and what was stripped. `.gitignore` blocks every audio extension.

Connections

- Audio Data Ethics and Handling · Consent Form - Audio Rate · Participant Consent Script and Form · Lab Session Runbook · Pipeline Input to Verdict · Synthesis/Risk Register rows 3, 14 · Project Guidelines · Proposal §4

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